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A Research Review on Generative AI Models for Cloud Based Intrusion Detection

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***Abstract*—Cloud computing delivers on-demand, internet-based resources with instant access, pay-per-use pricing, and flexible scalability to match user needs. But at the same time, despite these benefits, it is highly susceptible to dynamic and sophisticated cyber-attacks because it is exposed over the internet and contains serious security problems, particularly in detecting novel and new cyber-attacks. Traditional intrusion detection systems (IDS), which are signature-based and static rule-based, fail to detect new attacks and are ineffective in dynamic clouds. For enhancing cloud-based intrusion detection, this research compares various generative artificial intelligence models (GenAI): Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Generative Pre-trained Transformers (GPT) and diffusion-based models. Each model was compared with respect to computational efficiency, dataset flexibility, and performance metrics. The findings reveal that although GANs and VAEs have stronger recall, diffusion models and GPT have better accuracy and precision and are therefore more suitable in resource-limited or Internet of Things settings and beneficial in the detection of anomalous attacks. The findings reveal that no model is best all the time; rather, decisions need to be made based on the nature of threats, the nature of the dataset, and the strength of the system. Using the right generative models can make intrusion detection frameworks much more reliable, flexible, and able to find threats in dynamic cloud situations.**

Keywords—Cloud-based intrusion detection, generative AI, GANs, VAEs, GPT, Diffusion Models.

I. INTRODUCTION

Cloud computing has changed how businesses store data and run processes by giving them scalable technology that has also changed how businesses work in many other areas [1, 2, 3, 4]. Even though cloud systems are becoming more and more important for modern businesses, they also pose major security risks. Traditional intrusion detection systems depend on fixed rule sets or signatures that have already been set. This makes it challenging for them to detect new or unknown threats [5, 2, 6]. Because of this limitation, there are enormous concerns about how secure and effective cloud systems are, especially since cyber threats change to become intelligent rapidly.

The increasing complexity of security threats and the extensive use of the cloud need the development of security systems that are adaptable and learn from their experiences. Traditional anomaly detection methods, being mainly rule-based, are confronted with the complexity and dynamics of cyber threats [2, 5]. Therefore, researchers have been investigating new technologies, specifically generative models, to help tackle new security challenges in cloud infrastructures [8, 9, 10, 11].

Generative models such as GANs, VAEs, GPT and diffusion-based approaches are new data combining and augmentation methods that are highly promising for class-imbalance tasks as well as outlier detection tasks. The models are used to create synthetic samples that are near to actual attack data since there are no full cloud-based IDS datasets. GANs and VAEs are highly successful in anomaly detection due to the fact that they are trained in a manner such that they are able to detect patterns in normal as well as abnormal datasets. The incorporation of generative techniques into intrusion detection systems enables the detection of emerging threats, adaptation to changing threat patterns, and enhanced detection rates in cloud computing [13, 14]. GANs enrich training sets with artificial attack samples that make them robust and diverse. The technique sets data balance and forces machine learning models to learn optimally [5, 12]. The VAE system uses its compression ability to detect abnormal patterns in cloud traffic [11].

A. Aim and Scope

This study's objective is to investigate how generative modeling techniques can find gaps in cloud infrastructure even when there fails to be an extensive amount of training data, varied danger landscapes, and strange things constantly happen. Researchers are investigating more advanced generative modeling techniques, including generative adversarial networks, variational autoencoders, transformer-based models, and diffusion-based frameworks. This study examines cloud security app architectural patterns, pros and cons, and real-world uses [3, 5]. It examines the following areas in detail.

a) The theoretical foundations of each generative model.

b) Frameworks used in threat detection with generative AI.

c) Putting these concepts into action on cloud computing platforms.

d) The classification rate and positive predictive value (precision), true positive rate (recall), and F-measure (F1-Score) metrics will be used to keep track of how well these models are doing. [7, 12].

The remaining research component consists of four sections. The literature review on generative AI models is reviewed in the second section. The comparative architecture and computational assessment of GenAI

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models discussed in the third section. A comparative study is presented in the fourth section. The fifth part concludes the current study and suggests its future direction.

II. Literature Review

Cloud computing systems experience an escalating need for intrusion detection solutions because cyberattack complexity continues to rise. Traditional anomaly detection methods fail to identify both new and zero-day attacks because they cannot handle the complex nature of cloud data and unbalanced attack datasets. The challenges in IDS detection can be overcome through the use of advanced machine learning techniques and generative models including GANs, VAEs, GPT and Diffusion Models which enhance IDS performance and handle large data volumes effectively.

A. Generative Adversarial Networks (GANs)

The main challenge in intrusion detection comes from imbalanced datasets which GANs effectively address. GANs consist of two neural networks which perform a competitive training procedure to generate synthetic data. The generator produces artificial invasion examples while the discriminator attempts to distinguish between authentic and artificial data. The IDS becomes more effective because it receives artificial data for analysing uncommon attack methods. The classifier gains better performance in detecting unanticipated threats through this method. [2, 9]. The application of GANs has produced legitimate harmful traffic within cloud system environments. The training process of IDS models becomes more effective at detecting rare attacks which results in enhanced detection rates and improved protection against zero-day vulnerabilities. The application of GANs in IDS systems produces superior results compared to standard models particularly for detecting unusual and emerging threats that standard systems fail to detect. [6, 10].

B. Variational Autoencoders (VAEs)

The anomaly detection process becomes efficient through the use of VAEs that generate simplified latent representations of input data. VAEs function as generative models which aim to discover the fundamental distribution of data. The models have demonstrated successful representation of complex high-dimensional data characteristics in cloud systems. Intrusion detection through VAEs works by compressing input data into latent space to detect behavior anomalies [2]. The process of reconstructing input data enables the detection of anomalies by evaluating reconstruction errors. VAEs demonstrate strong performance in network traffic data simplification yet their latent representations sometimes exhibit reduced stability when compared to other methods. The Twin VAE (TVAE) and the Constrained Twin VAE (CTVAE) have improved VAE stability by implementing class separation constraints in the latent space. These models function better than others for cloud IDS system applications. These models lower the computational requirements and memory consumption thus making them appropriate for restricted resource environments found in clouds [11, 15].

C. Generative Pretrained Transformers (GPT)

The advancement of cybersecurity through GPT has led to better real-time threat identification and analysis. The GPT models gain recognition for their ability to produce writing which resembles human language while also understanding it. The current application of GPT models includes threat detection through network traffic and system log evaluation. The ability of GPT to analyze contextual data elements allows it to spot assault-related patterns. GPT models serve cloud security applications by processing large cloud data sets and user actions and network communications for immediate issue detection [7, 16]. The ability of GPT models to learn from new data continuously makes them highly beneficial for dynamic cloud environments. GPT models demonstrate both accurate threat detection and high recall rates thus becoming effective tools for proactive cloud infrastructure threat management according to research findings [7, 17].

D. Diffusion

The research community studies diffusion models because they produce new data while showing promise to handle unbalanced attack datasets IDS. The process of diffusion models involves adding noise to data in stages before developing the ability to transform this noise back into original data which expands training datasets. The application of diffusion models in cloud computing enables the creation of artificial data for uncommon attack types which addresses the class imbalance issue. Adding these produced samples to the dataset makes intrusion detection models stronger and better at finding attacks that are infrequent or have never been seen before [2, 18]. When combined with other methods like BiLSTM classifiers, diffusion models have been demonstrated to improve detection accuracy and make cloud IDS systems better at dealing with a wide range of attack situations [6, 18, 19].

III. Comparative Architecture and Computational Assessment of Generative Models

We provided a comprehensive analysis of prominent generative model architectures, emphasizing their foundational training procedures and mechanisms for synthetic data creation. Table I shows the main architecture, intended use, pros and cons of the most important generative models used in IDS. GANs are great at adding to datasets that aren't balanced [12], VAEs are great at compressing and reconstructing input data [11], GPT-based systems are great at finding contextual anomalies in logs [7] and diffusion models are great at creating rare attacks [6].

TABLE I. Architectural Comparison of Generative Models for Intrusion Detection

Model	Architecture	Main Application	Strength	Main Limitation	Data set	References
GAN	Generator and Discriminator	Anomaly Detection	Augments rare classes, reduces data imbalance	Computationally expensive, training instability	NSL-KDD, UNSW-NB15, IoT	[4, 12, 20, 21]
VAE	Encoder-	Dimensionality Reductio	Effective for feature	Latent space instabilit	NSL-KDD	[11, 15, 22]

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	Decoder	n, Anomaly Detection	compression in high-dimension data	y, less effective in multi-class	, HDFS, IoT	
GPT	Transformer	Real-time Threat Detection	High contextual understanding, learns from cloud logs	High computational load, large model size	CICIDS, 9018 Cyber Events	[7, 23]
Diffusion Model	Forward-Reverse Diffusion	Synthetic Data Generation	Strong in rare attack generation and balancing	Complex training and resource-intensive	BoT-IoT, ICPS	[6, 18]

Table II compares generative models on the quality and utility of synthetic data they generate. GANs and diffusion models deliver high-quality synthetic data, making them effective for handling imbalanced attack classes [12, 18]. VAEs contribute moderately by reconstructing anomalies, and GPT assists in real-time semantic understanding rather than data generation [23].

TABLE II. SYNTHETIC DATA GENERATION COMPARISON

Model	Synthetic Data Quality	Real-world Applicability	Impact on Detection Rate	References
GAN	High	High	Significantly increases rare attack detection	[20, 21]
VAE	Moderate	High	Improves detection of rare attacks	[11, 15]
GPT	Low	Moderate	Enhances contextual anomaly analysis	[7, 23]
Diffusion Model	High	High	Boosts performance for imbalanced data	[6, 18]

Table III compares the computational cost, training time, and model size for each approach that is helpful, especially in resource-constrained cloud environments. The effectiveness and expandability of generative models are shown in Table III. GPTs are very strong, but they need an extensive amount of resources to work, which makes it hard to use them in places with limited resources [23]. GANs and Diffusion Models also use an enormous amount of resources. On the other hand, VAEs achieve an optimal balance between performance and computational demand [22].

TABLE III. TRAINING TIME AND RESOURCE COMPARISON

Model	Computational Cost	Training Time	Model Size	References
GAN	High	Moderate-High	Large	[4,12,21]
VAE	Moderate	Moderate	Moderate	[11,15,22]
GPT	Very High	Long	Very Large	[7,23]
Diffusion Model	High	High	Moderate	[6, 18]

IV. COMPARATIVE ANALYSIS

Table IV compares these models for finding intrusions in a cloud-computing system [24]. Each model has used a different set of data and evaluated accordingly using several metrics. This group of methods includes diffusion-based methods and GPT-driven strategies, which show almost perfect detection results [7, 18], making them perfect for settings with a lot of changing threats. Despite having slightly lower performance, the GAN and VAE models are still competitive and can be used for edge and resource-limited applications [20, 22].

TABLE IV. PERFORMANCE METRICS OF GENERATIVE MODELS IN CLOUD IDS

Model	Dataset	Accuracy	Precision	Recall	F1-Score	Refere
GAN	UNSW-NB15	98%	96%	97%	95%	[12]
VAE Autoencoder + MLP	HDFS	97.50%	-	98%	97.00%	[22]
GPT-3	CICIDS, System Logs	99.90%	99%	99%	99%	[7]
Diffusion Model	Industrial Network	100%	99.70%	99.60%	98%	[18]

Fig. 1 displays that diffusion models and GPT give more exact and correct outcomes than GANs and VAEs. With a 98% competitive recall, VAEs reduced the number of false positives. It takes a lot of computing power to run diffusion models and GPT, but they are very good at finding things. GANs and VAEs, on the other hand, have balanced performance and less processing complexity.

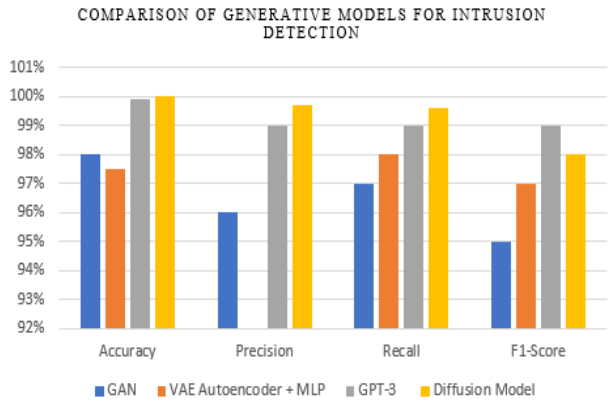


Fig.1. Performance Metrics of Generative Models for Cloud IDS

With their generator and discriminator parts, GANs successfully fix data imbalance by creating synthetic

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attacks. They do, however, need a lot of computing power, and training them can be hard. VAEs are great at reducing the number of dimensions and finding anomalies while using little memory. However, their hidden representations may become unstable in situations with more than one class.

With its transformer architecture, GPT can find cyber risks in real time by correctly analyzing complex threat patterns. But it has problems because it needs a lot of computing power and the model is very big. Diffusion models handle uneven data well by making fake attack samples for uncommon patterns, but they are very computationally expensive and need to be processed in several steps..

V. Conclusions

This study compared GANs, VAEs, GPT, and diffusion models for cloud intrusion detection. Diffusion models and GPT detect complex threats with more accuracy and precision. However, they demand high computational power.

GANs and VAEs provide greater computational efficiency and recall. VAEs detect anomalies with little memory consumption, while GANs address data imbalance with synthetic attack generation. None of these models is ideal. Threat, resources, and dataset properties determine the option. Future work should investigate hybrid architectures and federated learning to balance detection accuracy and computing efficiency in dynamic cloud environments.

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